Exam: Laws of Exponents & Radicals

Instructions: Simplify completely. Write final answers with *positive* exponents or in *simplest radical form*. Show all steps.

Part A. Laws of Exponents

1) $\dfrac{(x^6)^{\frac{1}{2}}}{x^{\frac{3}{2}}}$

2) $\left(\dfrac{x^9}{x^3}\right)^{\frac{2}{3}}$

3) $\dfrac{x^{\frac{5}{2}} \cdot x^{\frac{3}{4}}}{x^{\frac{1}{4}}}$

4) $\left(x^{\frac{2}{5}}\right)^{\frac{5}{2}} \cdot x^{-1}$

5) $\dfrac{(x^{12})^{\frac{1}{4}}}{x^{\frac{5}{2}}}$

Part B. Simplification of Radicals

6) $3\sqrt{18} + 2\sqrt{8}$

7) $\sqrt{50} - \sqrt{8} + \sqrt{18}$

8) $2\sqrt[3]{16} + 5\sqrt[3]{54}$

9) $4\sqrt{75} - 3\sqrt{12} + 2\sqrt{48}$

10) $\sqrt[3]{128} - 2\sqrt[3]{54} + \sqrt[3]{250}$

Part C. Laws of Radicals

11) $\sqrt{x^8} \cdot \sqrt[3]{x^3}$

12) $\dfrac{\sqrt[4]{x^{10}}}{\sqrt{x}}$

13) $\sqrt{x} \cdot \sqrt[3]{x^2}$

14) $\dfrac{\sqrt[3]{x^5} \cdot \sqrt{x^3}}{\sqrt[6]{x}}$

15) $\dfrac{\sqrt[5]{x^{12}}}{\sqrt[10]{x}}$

Solutions
Part A

1) $\frac{(x^6)^{\frac{1}{2}}}{x^{\frac{3}{2}}} = \frac{x^{6\cdot\frac{1}{2}}}{x^{\frac{3}{2}}} = \frac{x^3}{x^{\frac{3}{2}}} = x^{3-\frac{3}{2}} = x^{\frac{3}{2}}$.

2) $\left(\frac{x^9}{x^3}\right)^{\frac{2}{3}} = \left(x^{9-3}\right)^{\frac{2}{3}} = (x^6)^{\frac{2}{3}} = x^{6\cdot\frac{2}{3}} = x^4$.

3) $\frac{x^{\frac{5}{2}}\cdot x^{\frac{3}{4}}}{x^{\frac{1}{4}}} = x^{\frac{5}{2}+\frac{3}{4}-\frac{1}{4}} = x^{\frac{5}{2}+\frac{2}{4}} = x^{\frac{5}{2}+\frac{1}{2}} = x^3$.

4) $\left(x^{\frac{2}{5}}\right)^{\frac{5}{2}} \cdot x^{-1} = x^{\left(\frac{2}{5}\right)\left(\frac{5}{2}\right)} \cdot x^{-1} = x^1 \cdot x^{-1} = x^0 = 1$.

5) $\frac{(x^{12})^{\frac{1}{4}}}{x^{\frac{5}{2}}} = \frac{x^{12\cdot\frac{1}{4}}}{x^{\frac{5}{2}}} = \frac{x^3}{x^{\frac{5}{2}}} = x^{3-\frac{5}{2}} = x^{\frac{1}{2}}$. (Equivalently, $\sqrt{x}$.)

Part B

6) $3\sqrt{18} + 2\sqrt{8} = 3\sqrt{9\cdot 2} + 2\sqrt{4\cdot 2} = 3\cdot 3\sqrt{2} + 2\cdot 2\sqrt{2} = 9\sqrt{2} + 4\sqrt{2} = 13\sqrt{2}$.

7) $\sqrt{50} - \sqrt{8} + \sqrt{18} = \sqrt{25\cdot 2} - \sqrt{4\cdot 2} + \sqrt{9\cdot 2} = 5\sqrt{2} - 2\sqrt{2} + 3\sqrt{2} = 6\sqrt{2}$.

8) $2\sqrt[3]{16} + 5\sqrt[3]{54} = 2\sqrt[3]{8\cdot 2} + 5\sqrt[3]{27\cdot 2} = 2\cdot 2\sqrt[3]{2} + 5\cdot 3\sqrt[3]{2} = 4\sqrt[3]{2} + 15\sqrt[3]{2} = 19\sqrt[3]{2}$.

9) $4\sqrt{75} - 3\sqrt{12} + 2\sqrt{48} = 4\sqrt{25\cdot 3} - 3\sqrt{4\cdot 3} + 2\sqrt{16\cdot 3} = 4\cdot 5\sqrt{3} - 3\cdot 2\sqrt{3} + 2\cdot 4\sqrt{3} = 20\sqrt{3} - 6\sqrt{3} + 8\sqrt{3} = 22\sqrt{3}$.

10) $\sqrt[3]{128} - 2\sqrt[3]{54} + \sqrt[3]{250} = \sqrt[3]{64\cdot 2} - 2\sqrt[3]{27\cdot 2} + \sqrt[3]{125\cdot 2} = 4\sqrt[3]{2} - 2\cdot 3\sqrt[3]{2} + 5\sqrt[3]{2} = (4-6+5)\sqrt[3]{2} = 3\sqrt[3]{2}$.

Part C

11) $\sqrt{x^8}\cdot\sqrt[3]{x^3} = x^{8/2}\cdot x^{3/3} = x^4\cdot x^1 = x^5$.

12) $\frac{\sqrt[4]{x^{10}}}{\sqrt{x}} = \frac{x^{10/4}}{x^{1/2}} = x^{\frac{10}{4}-\frac{1}{2}} = x^{\frac{5}{2}-\frac{1}{2}} = x^2$.

13) $\sqrt{x}\cdot\sqrt[3]{x^2} = x^{1/2}\cdot x^{2/3} = x^{\frac{1}{2}+\frac{2}{3}} = x^{\frac{7}{6}}$

14) $\frac{\sqrt[3]{x^5}\cdot\sqrt{x^3}}{\sqrt[6]{x}} = \frac{x^{5/3}\cdot x^{3/2}}{x^{1/6}} = x^{\frac{5}{3}+\frac{3}{2}-\frac{1}{6}} = x^{\frac{10}{6}+\frac{9}{6}-\frac{1}{6}} = x^{\frac{18}{6}} = x^3$.

15) $\frac{\sqrt[5]{x^{12}}}{\sqrt[10]{x}} = \frac{x^{12/5}}{x^{1/10}} = x^{\frac{12}{5}-\frac{1}{10}} = x^{\frac{24}{10}-\frac{1}{10}} = x^{\frac{23}{10}}$